Frank Fan

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SUMMARY

PhD student in computer graphics at the University of Waterloo working on predictive rendering and biophysically based material appearance modelling, with a peer-reviewed publication at the Eurographics Symposium on Rendering (EGSR) 2026 and a Master's thesis introducing a multispectral framework for simulating chlorophyll fluorescence. Complemented by four software engineering co-op terms building production C++ and C# systems for 3D simulation and data visualization, spanning ray tracing, BRDF and microfacet modelling, radiometry, spatial acceleration structures (BVH), OpenGL pipelines, and performance optimization on meshes of approximately one million triangles. Seeking a research internship in rendering and appearance modelling.

TECHNICAL SKILLS

Programming Languages: C++, C, C#, Python, GLSL, Java, JavaScript, SQL, Bash, Scala, MATLAB, PHP
Graphics and Rendering: Physically based rendering, predictive and biophysically based appearance modelling, spectral rendering, fluorescence simulation, radiometry, light transport, BRDF and microfacet models (GGX), ray tracing, path tracing, rasterization, Monte Carlo integration, OpenGL, GLSL shaders, texture and bump mapping, mesh processing
Algorithms and Methods: Spatial data structures, bounding volume hierarchies (BVH), collision detection, computational geometry, numerical methods, stochastic simulation, performance profiling and optimization
Tools and Platforms: Git, Perforce, Subversion, Linux, .NET, WinForms, ImGui, devDept Eyeshot, PostgreSQL, CI/CD automation, LaTeX

EDUCATION

Doctor of Philosophy (PhD), Computer Science | University of Waterloo, Waterloo, ON | Sep 2026 – Present
Research advisor: Prof. Gladimir V. G. Baranoski, Natural Phenomena Simulation Group.
Research area: physically based rendering, light transport, predictive appearance modelling.

Master of Mathematics (MMath), Computer Science | University of Waterloo, Waterloo, ON | Sep 2024 – Apr 2026
Research advisor: Prof. Gladimir V. G. Baranoski, Natural Phenomena Simulation Group. GPA: 4.0/4.0.
Thesis: "A Biophysically-Based Framework for the Simulation and Visualization of Chlorophyll Fluorescence Under Different Illumination Conditions." hdl.handle.net/10012/23104

Bachelor of Computer Science (BCS) | University of Waterloo, Waterloo, ON | Sep 2019 – Apr 2024
GPA: 3.8/4.0. Relevant coursework: CS488 Computer Graphics, CS489 Advanced Topics in Computer Graphics, CS370 Numerical Computation, CS341 Algorithms, CS343 Concurrent and Parallel Programming.

PUBLICATIONS

- Sun, Y., G. V. G. Baranoski, **F. Fan**, and S. Steinfield. "A Phasor-based Camera-Space Construction for Making Radiometric Accumulation Assumptions Explicit." *Eurographics Symposium on Rendering (EGSR) 2026*, Symposium Track, Eurographics Association, Bordeaux, France, July 2026. DOI: 10.2312/sr.20261012
- **Fan, F.**, and G. V. G. Baranoski. "A Study on Rendering Techniques to Visually Represent Sparkles." Technical Report CS-2024-02, University of Waterloo, 2024. npsg.uwaterloo.ca/resources/docs/tr/Report-Sparkles_CS-2024-02.pdf

PROFESSIONAL EXPERIENCE

Lead Teaching Assistant, Computer Graphics | University of Waterloo, Waterloo, ON | Sep 2024 – Apr 2026

- Lead teaching assistant for CS488 Introduction to Computer Graphics, supporting approximately 50 students per term on ray tracing, rasterization pipelines, transformation hierarchies, and shader debugging.
- Coordinated a team of 3 teaching assistants across grading and office hours, standardizing rubrics for 250 assignments and final rendering projects.
- Delivered technical support across CS488, CS370 (Numerical Computation), and CS245 (Logic and Computation) through office hours and Piazza, resolving approximately 150 student questions per term.

Graduate Researcher, Physically Based Rendering | Natural Phenomena Simulation Group, University of Waterloo | Sep 2024 – Present

- Co-authored "A Phasor-based Camera-Space Construction for Making Radiometric Accumulation Assumptions Explicit," published at the Eurographics Symposium on Rendering (EGSR) 2026.
- Developed a biophysically based framework for simulating and visualizing chlorophyll fluorescence under ultraviolet and visible excitation, grounded in fluorescence spectroscopy principles and generalizable to other fluorescent materials (Master's thesis, University of Waterloo, 2026).

- Validated the framework's predictive capability against qualitative traits observed in physical measurements, and demonstrated its use in realistic image synthesis by rendering chlorophyll solutions across varied experimental and illumination conditions.
- Maintain the group's public research web presence (npsg.uwaterloo.ca) on Linux, publishing technical reports, datasets, and figure resources used by the wider rendering research community.

Undergraduate Research Assistant | University of Waterloo, Waterloo, ON | Jan 2024 – Apr 2024

- Researched the Uniform Theory of Diffraction (UTD) and GGX microfacet models under Prof. Shlomi Steinberg to represent wave-optics effects in physically based BRDFs.
- Analyzed the trade-offs between wave-optical and geometric-optics reflectance models for real-time and offline rendering applications.

Undergraduate Research Assistant | University of Waterloo, Waterloo, ON | Sep 2023 – Dec 2023

- Conducted a comparative study of stochastic, spectral, and rasterization-based techniques for rendering sparkle optical phenomena, under Prof. Gladimir Baranoski.
- Authored a technical report (CS-2024-02) published by the University of Waterloo School of Computer Science, evaluating several rendering approaches on visual fidelity and computational cost.

Software Developer | Rocscience Inc., Toronto, ON | Jan 2023 – Apr 2023

- Developed C++ algorithms to compute rockfall simulation results, including trajectory height, impact angle, kinetic energy, and velocity, shipped in RocFall3, a commercial geotechnical analysis product.
- Designed a spatial indexing data structure to store and query impact events across large-scale meshes of approximately one million triangles and hundreds of objects, reducing query time by approximately 20%.
- Built interactive 3D visualizations of rockfall-barrier impact data using OpenGL through the devDept Eyeshot framework, including custom mesh rendering and color-mapped energy heatmaps.

Software Developer | Tactic Studios, Toronto, ON | May 2022 – Aug 2022

- Built a continuous integration pipeline against the Perforce API to synchronize feature branches, trigger remote builds, and automate game asset baking, eliminating approximately 2 hours of manual work per week for a team of 20 developers.

Software Developer | Rocscience Inc., Toronto, ON | Sep 2021 – Dec 2021

- Rearchitected the spatial partitioning system for dense polygon meshes of approximately one million triangles around an optimized bounding volume hierarchy (BVH), accelerating point-mesh collision detection by approximately 1.25x.
- Reduced traversal complexity and improved cache locality in point-mesh collision queries, cutting simulation runtime from approximately 1.5 to 1.25 on production models.

Software Developer | PSI Technologies Inc., Saskatoon, SK | Jan 2021 – Apr 2021

- Developed C# features for a .NET WinForms desktop application using devDept Eyeshot, including an interactive 3D cross-section tool allowing users to clip a mesh along an arbitrary plane.
- Implemented a real-time mesh clipping algorithm that generates cross-section geometry on the fly to expose interior structure, used by engineering firms.

PROJECTS

Stochastic Colloidal Particle Collision Simulator | MATLAB, Monte Carlo methods | 2023

- Implemented a physically based model of Brownian motion in colloidal particle systems, using stochastic techniques and Monte Carlo integration to detect and resolve inter-particle collisions.

Recursive Ray Tracer | C++, OpenGL | 2022

- Built an extended recursive ray tracer supporting reflection, refraction, glossy reflection and refraction, texture mapping, bump mapping, and stochastic anti-aliasing.